

Prashant Panta

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EDUCATION

Texas State University—GPA: 3.92

Exp. Graduation: May 2027

Computer Science (Honors), Ingram School of Engineering

Related Coursework: Discrete Math I & II, Calculus I, II & III, Data Structures & Algorithms, Assembly Language, Unity XR, Applied Linear Algebra, Probability & Statistics, Object-Oriented Programming, Research Ethics and Methodologies, Parallel Computing, System Security

PUBLICATIONS

- Ahmed, T., Chen, H., & **Panta, P.** (2024). Human Motion Gait Analysis Using IMU Sensors. IEEE 14th International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER), 714–719. <https://doi.org/10.1109/cyber63482.2024.10748719>
- **Panta, P.** (2025). Applications of Wearable Sensors for Gait Analysis. Texas State Undergraduate Research Journal, 13(i). <https://doi.org/10.58997/7bhxxm11>
- Ahmed, T., Panta, P., & Chen, H. (2025). A Review of Smart Collars and Innovations in Behavior and Health Monitoring Technologies for Cattle Farming. Preprint. ResearchGate.
- Pashupatinath—research eBook. (2023, June 26). Project Pashupatinath. <https://pashupatinath.reinstallinghope.org/>

RESEARCH EXPERIENCE

High Performance Computing Research Internship, TXST Division of IT

Jan 2025 – Present

- Contributing to development, benchmarking, and optimization of Texas State's LEAP2 HPC cluster, documenting GPU acceleration workflows and optimizing job scheduling policies for research applications
- Designing new Vice President of IT office internship structure for upcoming cohorts (30–40 interns), developing technical onboarding materials, training workflows, and project frameworks

Lehigh University, NSF RTA Program, Summer Research Intern

Jun 2025 – Aug 2025

- Developed an autonomous underwater ROV using PID control on a Pixhawk flight controller, integrating six-degree-of-freedom thruster control on an NVIDIA Jetson Nano.
- **Awarded 2nd** place out of 103 projects at the Lehigh University Summer Research Symposium

TXST CIVS Lab || Undergraduate Researcher

Sep 2024 – Apr 2025

- Analyzed muscle fatigue and movement patterns using electromyography (EMG), motion sensors (IMU), and foot-pressure sensors to support construction worker safety research
- Led tutoring sessions on ergonomics, gait analysis, and workplace safety for 50+ students in collaboration with engineering faculty

EXPERIENCE & LEADERSHIP

Chief Information Officer Student Advisor || Texas State University

Aug 2025 – Present

- Selected as one of two students to provide strategic advisement to university CIO and Vice President of IT on technology initiatives affecting 50,000+ students and faculty
- Led improvements to IP Address Roaming Architecture, increasing online accessibility; provided advisement on Wi-Fi routing lifecycle issues, leading to \$1.2M network infrastructure upgrade

Director of Community Outreach || Student Government

Aug 2025 – May 2026

- Created the university's first International students role within community outreach, establishing formal representation for 1,500+ international students
- Serve as liaison between international students, student government, and university administration

Presidential Ambassadors || Gold Star Society

Aug 2025 – Present

- Represent the Office of the President at official Texas State University events, fostering connections between university leadership, students, and distinguished guests
- Serve as a student ambassador and role model, embodying Texas State's values of excellence, leadership, and service

Vice President/President || Nepalese Student Association

Apr 2025 – April 2026

- Representing 470+ Nepali students; preparing to support cultural events, student engagement efforts, and international student advocacy through upcoming initiatives and collaborative leadership

PROJECTS

HackMIT 2025 Finalist (Top 10/360+ Projects, 1100 Participants)

Sep 2025

- Built Bagaicha, a smart agriculture platform integrating 7-in-1 soil sensors and a web dashboard for real-time plant health monitoring
- Implemented a computer vision pipeline for plant species classification (92% accuracy) and personalized care recommendations

Autonomous F1TENTH Racecar || Lehigh Summer Internship

Jun 2025– Aug 2025

- Revived 5+ F1TENTH autonomous racecars that had been inactive for over two years; migrated control systems from ROS1 to ROS2
- Refined PWM-based steering control logic by improving lap times by 18%, resulting in smoother and stable turning

HONORS & AWARDS

- William Hatfield Hogue Memorial Endowed Scholarship – Highest Leadership Award, TXST
- Gold Award, Most Spirited Student Leader – Maroon and Gold Awards, Texas State University
- Overall Balanced Man Scholarship—First recipient, Sigma Phi Epsilon, Texas State University
- Undergraduate Computer Science Excellence Award – Texas State University, 2026
- Eugene Schurg Endowed Scholarship—Sigma Phi Epsilon · National STEM Scholarship 2026
- Suzanne B. Patenaude Endowed Scholarship – Texas State University · 2026
- Jeffrey A. Slomka Memorial Scholarship in Computer Science – Texas State University · 2026
- Texas State University Support Foundation Endowment · 2026
- HackMIT Top 10 Finalist – 1,100+ participants · 2025
- 2nd Place, Lehigh Summer Research Symposium – 103 competing projects · 2025

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, SQL, LaTeX

Machine Learning & Data Science: TensorFlow, Scikit-learn, PyTorch, OpenCV, NumPy, Pandas, Matplotlib

Robotics & Autonomous Systems: ROS2, MAVLink, ArduPilot, Gazebo, RViz, QGroundControl, Linux

High-Performance Computing: SLURM job scheduling, parallel processing, cluster computing

Development Tools: Git, GitHub, Bitbucket, MATLAB, Jira, Confluence